UniWorld: A Unified World Model for Real-Time Visual World Simulation

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Continuous visual world simulation requires coupling evolving scene geometry with temporally persistent appearance. We present UniWorld, a unified framework that integrates depth trajectories and reference appearance within a shared video generation model. The central design maintains visual and geometric histories on a common timeline. Noise-aligned overlapping continuation places generated history at the prescribed sampling noise level and commits only newly generated latents. Rolling depth latent state preserves the corresponding geometric controls, incrementally reads newly available geometry, and keeps the conditioning fixed within each sampling window. Together, these mechanisms ensure exact preservation of latent history while bounding the active depth state by the window size. We evaluate continuous generation, paired history interventions, and long-horizon behavior across diverse scene categories. The same generator supports estimated, native, and on-demand simulator depth, while cached and incremental execution paths produce bitwise-identical outputs. Appearance changes and geometric control branches operate through the same interface without modifying committed history. The measured steady-state DiT execution path achieves real-time generation throughput, demonstrating the efficiency of UniWorld for continuous visual world simulation. Project page: https://zephipage.github.io/UniWorld.

Keywords: World models; Video generation; Depth conditioning; Continuous visual simulation



Figure 1 UniWorld: from world geometry to continuous visual observations. Conceptual overview: depth trajectories describe geometric evolution, while reference RGB specifies appearance for a shared video renderer.

1 Introduction

World models connect observations with environment dynamics, supporting simulation, prediction, and decision-making [Ha and Schmidhuber, 2018, Bruce et al., 2024, Zhu et al., 2026]. Graphics engines and physical simulators provide structured and controllable scene states, whereas video generators provide rich visual appearance. Bridging these capabilities requires a renderer that can continuously transform evolving geometry into temporally consistent visual observations. We focus on this simulator-to-renderer component: an upstream world source specifies a geometric trajectory, and a generative model produces the corresponding visual realization.

Different world sources expose distinct state variables, camera models, and control interfaces, which makes direct coupling to a shared visual generator difficult. We observe that the visible geometry produced by these sources can instead be represented in a common image-space form: depth. Image-aligned depth captures scene layout, occlusion, and geometric changes, while a reference RGB image specifies texture, color, and appearance semantics. This separation maps heterogeneous world states to a common rendering interface, enabling the same generator to operate on recorded scenes, simulator trajectories, and on-demand geometric inputs.

Continuous rendering introduces a second challenge: generation must remain consistent across finite sampling windows. Long-video methods extend temporal context, use overlapping windows, or compress history [Henschel et al., 2024, Zhang et al., 2025, Gao et al., 2025a], while autoregressive approaches condition future generation on earlier observations [Huang et al., 2025, Zhu et al., 2026]. Depth-driven rendering must additionally preserve the geometric conditions associated with those observations. Continuous generation therefore requires two synchronized histories: the generated visual state and the depth state that produced it. Previously generated observations must remain fixed, the associated depth values must remain unchanged, and only newly requested geometry should produce newly committed outputs. A shared temporal contract makes these state transitions explicit and testable.

We propose UniWorld, a unified generative rendering framework that maintains visual and geometric states on a common timeline. *Noise-aligned overlapping continuation* places previously generated latents in the prefix of the next window and constrains them to the corresponding flow time throughout sampling. *Rolling depth latent state* preserves the matching geometric history, incrementally reads only newly available depth positions, and keeps the conditioning fixed within each window. After sampling, the system commits only the newly generated video suffix. The sampling window can therefore advance without rewriting committed latents or altering the controls associated with the overlapping region.

This design bounds the active state and per-window computation independently of the total sequence duration, thereby enabling continuous long-horizon rendering. We verify the state contract through actual model executions and evaluate UniWorld across four scene categories using estimated and native depth, controlled fixed-window interventions, and extended trajectories. An on-demand engine further demonstrates reference-appearance changes and branch-conditioned continuation through the same interface. The measured steady-state DiT execution path achieves real-time rendering throughput while maintaining a bounded active state during continuous generation.

Our contributions are:

1. **UniWorld.** A unified visual simulation framework that represents heterogeneous geometric trajectories using image-aligned depth and combines them with reference appearance, providing a common generative rendering interface for recorded scenes, simulators, and on-demand world trajectories.

2. **Noise-aligned overlapping continuation.** A sampling rule that constrains a generated multi-frame prefix at every flow time and commits only the newly generated suffix, thereby preserving previously generated latents while retaining temporal context for continued generation.

3. **Rolling depth latent state.** A synchronized geometric state that incrementally reads only newly available depth positions, preserves the conditions associated with committed observations, and maintains an active-state bound that is independent of sequence duration.

2 Related Work

2.1 Visual World Models

World Models learn compact representations of environment dynamics [Ha and Schmidhuber, 2018], and Genie learns interactive visual environments from video [Bruce et al., 2024]. Astra combines autoregressive denoising, noisy histories, and action adapters to support exploration, manipulation, and driving [Zhu et al., 2026]. These approaches link observations to actions or latent controls. UniWorld complements these approaches by providing a shared visual rendering interface: an upstream simulator supplies the evolving geometry, and the generator continuously renders the corresponding appearance.

2.2 Conditional Video Generation

Large video models support conditioning on text and images [Wan Team, 2025, MiniMax, 2026]. ControlNet, SparseCtrl, and CameraCtrl introduce structural or camera constraints [Zhang et al., 2023, Guo et al., 2023, He et al., 2024]. LongVie and LongVie 2 combine multiple control signals for long-video synthesis [Gao et al., 2025a,b]. Our implementation builds on H3, Video DeltaNet (VDN), Qwen3-VL, and the VideoX-Fun Union control branch [MiniMax, 2026, Xi et al., 2026, Bai et al., 2025, aigc-apps, 2026]. The original image provides persistent semantic context, while a generated boundary image and temporally aligned depth serve as local conditioning inputs. Our modifications focus on how these conditioning inputs and the generated history are retained across windows.

2.3 Long-Video Generation

StreamingT2V combines short- and long-term appearance memory [Henschel et al., 2024], and FramePack compresses frame context [Zhang et al., 2025]. Diffusion Forcing uses per-token noise levels during training, and Self Forcing addresses the mismatch between training histories and model-generated histories [Chen et al., 2024, Huang et al., 2025]. RePaint conditions diffusion sampling on known regions by replacing noisy observations [Lugmayr et al., 2022]. LongVie uses unified noise initialization and global control normalization [Gao et al., 2025a]. Building on these principles, we focus on a joint temporal contract for a flow renderer that uses eight evaluations. Under this contract, the multi-frame target prefix is matched to the current sampling noise level, canonical latent values are preserved in overlapping depth segments, and incremental output is aligned with the global VAE phase.

3 Approach

3.1 Task Definition and Preliminaries

The method takes as input a single reference image I_{ref}, optional text p, and a spatially and temporally aligned depth sequence D, and outputs a video $\hat{V}$ at playback rate f. The target generation rate for real-time operation is 14 FPS. For subsequent windows, boundary images are obtained from previously generated content. All temporal intervals use zero-based, half-open indexing. Let N and F denote the aligned temporal lengths in pixel space and latent space, respectively. Window i spans $\Omega_i = [a_i, b_i)$ in latent time, with length $L_i = b_i - a_i$, inherited history H_i of length h_i, depth condition C_i, boundary image I_i, and a slice ϵ_i of the global noise. The first window has $h_0 = 0$.

$$c_{\text{g}} = E_{\text{Qwen}}(I_{\text{ref}}, p), \qquad \hat{Z}_i = G_\theta^{(K)}(H_i, C_i, I_i, c_{\text{g}}, \epsilon_i), \qquad K = 8. \tag{1}$$

Here $G_\theta^{(K)}$ includes the projection and sampler described below. The windows are generated sequentially because each window depends on previously generated history, whereas generation within each window remains bidirectional.

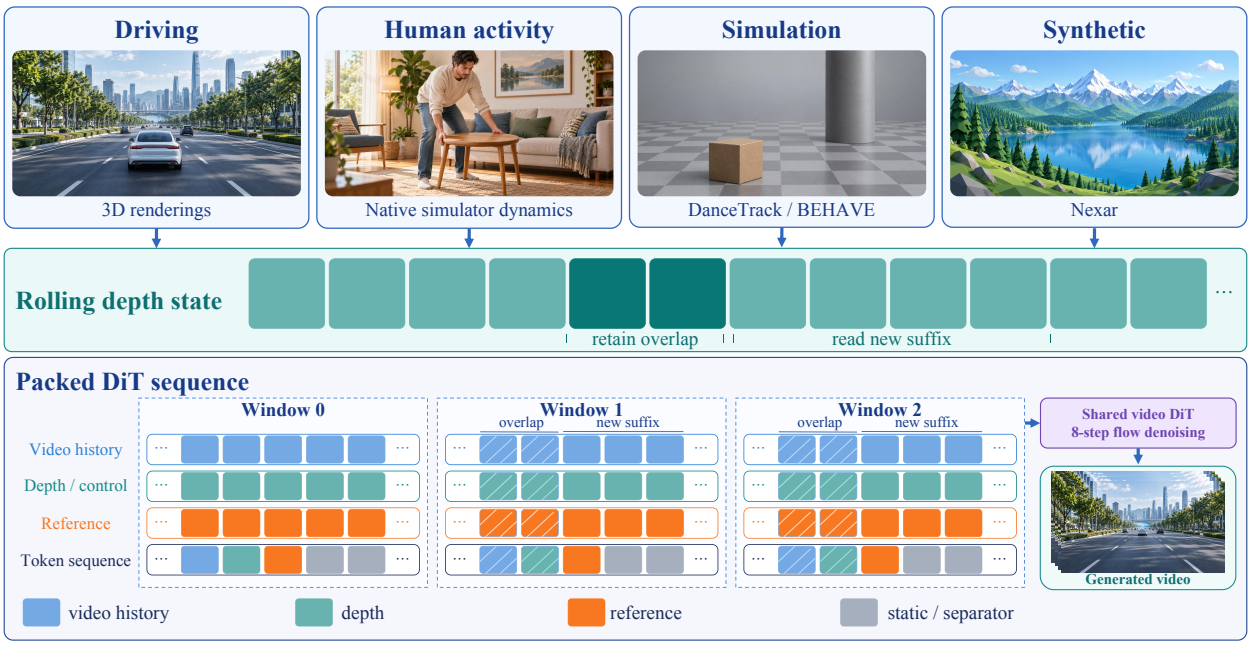


Figure 2 UniWorld: a unified renderer for heterogeneous world sources. Driving, human-activity, simulation, and synthetic sources are converted into a common rolling depth state. Each continuation window packs video history, depth/control, reference, and separator tokens into a shared DiT sequence. Overlapping positions are retained, only the newly required depth suffix is read, and only the newly generated video suffix is committed to the output stream.

3.2 UniWorld: A Unified Framework for Visual World Simulation

For source m, let $\xi^{(m)}$ denote its state or observations, and let $\kappa^{(m)}$ denote the corresponding camera and alignment information. A source adapter $\mathcal{A}_m$ produces visible depth, while $\mathcal{N}_m$ standardizes its representation:

$$\bar{D}^{(m)} = \mathcal{N}_m\Big(\mathcal{A}_m(\xi^{(m)}, \kappa^{(m)})\Big), \qquad C = \mathrm{Patchify}\big[\mathrm{Norm}_z\big(E_{\mathrm{VAE}}(\mathrm{RGBReplicate}(\bar{D}))\big)\big]. \tag{2}$$

Time, viewpoint, crop, near/far direction, scale, and encoding phase remain consistent throughout the trajectory. We replicate grayscale depth across the RGB channels, use the VAE posterior mode with FP32 latent normalization, and patchify with shape $(1, 2, 2)$. Each latent time position contains $S = (H_{\mathrm{px}}/32)(W_{\mathrm{px}}/32)$ spatial tokens, each with dimension 96. Both estimated and native depth follow the verified far-bright convention of the checkpoint. An online adapter encodes newly requested geometry under the same global phase; Appendix C.2 specifies the source mappings.

Appearance conditioning is provided through two paths (Figure 2). The original image and text are encoded once by Qwen3-VL and reused throughout the trajectory. A separate VAE first-frame condition uses $I_0 = I_{\mathrm{ref}}$ initially and generated RGB at the start of each subsequent window. These independent conditioning rows are distinct from the multi-frame history, which occupies the target video sequence itself.

3.3 Noise-Aligned Overlapping Continuation

History as a sampling boundary condition. We adopt the noise-to-data flow convention [Lipman et al., 2023]:

$$z_t = t z_{\mathrm{clean}} + (1-t)\epsilon, \qquad v = z_{\mathrm{clean}} - \epsilon, \qquad 0 \le t \le 1. \tag{3}$$

For window i, let M_i mask the first h_i latent positions corresponding to inherited history, and extend H_i by zeros outside this prefix. We impose the history through the projection

$$\boxed{\mathcal{P}_{i,t}(z) = M_i \odot \big[tH_i + (1-t)\epsilon_i\big] + (1 - M_i) \odot z.} \tag{4}$$

Thus, the inherited prefix follows the same noise-to-data trajectory as the current sample instead of being inserted as a clean latent into a noisy state. Global positions shared by overlapping windows use the same realization of global noise.

On the shifted flow grid $0 = t_0 < \cdots < t_K = 1$, we initialize

$$z_i^{(0)} = \mathcal{P}_{i,t_0}(\epsilon_i), \tag{5}$$

and perform

$$\tilde{z}_i^{(k+1)} = z_i^{(k)} + (t_{k+1} - t_k) v_\theta\left(z_i^{(k)}, t_k; \mathcal{C}_i\right), \qquad z_i^{(k+1)} = \mathcal{P}_{i,t_{k+1}}\left(\tilde{z}_i^{(k+1)}\right), \tag{6}$$

where

$$\mathcal{C}_i = (C_i, I_i, c_{\mathrm{g}}) \tag{7}$$

collects the fixed depth, boundary-image, and global appearance conditions for window i.

The projection ensures that the inherited prefix is aligned with the current flow noise level before every model evaluation. After each numerical update, the predicted prefix is replaced by its prescribed history trajectory, whereas the predicted suffix is retained. Consequently, the inherited history fully participates in the bidirectional forward pass within each window and guides newly generated content without drifting. At $t_K = 1$, the projected prefix exactly recovers H_i. Because the projection is an element-wise replacement, it requires no additional model evaluations.

History preservation and output commitment. For fixed history, mask, and noise, the projection is idempotent. The initialization and equation 6 imply

$$\mathcal{P}_{i,t}(\mathcal{P}_{i,t}(z)) = \mathcal{P}_{i,t}(z), \qquad M_i \odot z_i^{(k)} = M_i \odot [t_k H_i + (1 - t_k)\epsilon_i]. \tag{8}$$

Thus, the terminal prefix equals H_i. We preserve the previously committed sequence $\hat{Z}_{<i}$ and append only the newly generated suffix:

$$\hat{Z}_{\le i} = \mathrm{concat}\big(\hat{Z}_{<i}, z_i^{(K)}[h_i : L_i]\big). \tag{9}$$

Appendix A proves exact latent-history preservation. Section 4.4 evaluates the generated suffix under fixed-window interventions, while the decoder described below preserves the global phase during RGB commitment.

Window geometry and boundary images. Under the temporal geometry of the video VAE, $N = 17n + 5$ pixel frames correspond to $F = 5n + 2$ latent frames. The default pixel window, overlap, and stride are $(W, O, \Delta) = (345, 39, 306)$, corresponding to $(L_{\max}, h, \Delta_{\mathrm{latent}}) = (102, 12, 90)$. For the evaluation stream at 24 FPS, a 30-second request is internally padded to 736 pixel frames and 217 latents, and the decoded output is trimmed to 720 frames. The corresponding latent windows are $[0, 102)$, $[90, 192)$, and $[180, 217)$ (Figure 3). The subsequent windows start at pixel frames 306 and 612, while the nominal new-suffix boundaries are 345 and 651.

The next boundary RGB is taken at the *start of the next window*: decoding latents $[85, 97)$ and selecting local pixel frame 17 recovers frame 306, matching full-window decoding. RGB export preserves a single global VAE phase. Incremental delivery retains five unfinished blend frames and emits $[0, 340)$, $[340, 646)$, and then $[646, 720)$; these pixel-delivery boundaries differ from the latent commitment boundaries.

3.4 Rolling Depth Latent State

For $i > 0$, let $h_i = b_{i-1} - a_i$ denote the overlap with the preceding window. Rather than reprocessing overlapping geometry, we retain the corresponding canonical depth latents and append only newly requested positions:

$$\boxed{C_i = \mathrm{concat}\left(C_{i-1}[a_i - a_{i-1} : L_{i-1}],\ \mathrm{ReadDepth}(b_{i-1}, b_i)\right).} \tag{10}$$

The first window reads $[a_0, b_0)$. A state update is committed only after interval, shape, dtype, and finite-value checks pass; retrying a window that has already been committed returns the cached condition. Once constructed, the depth state remains fixed throughout all K model evaluations:

$$C_i^{(0)} = C_i^{(1)} = \cdots = C_i^{(K-1)} = C_i. \tag{11}$$

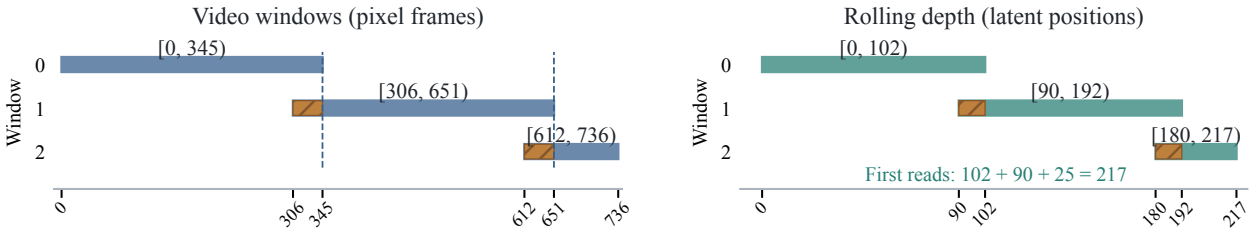


Figure 3 Aligned visual and geometric state. Hatched intervals denote inherited history. Depth is read exactly once over $[0, 102)$, $[102, 192)$, and $[192, 217)$. Dashed pixel lines mark the nominal latent-commitment boundaries, while stable incremental pixel output retains an additional five-frame blend tail.

Thus, unlike video history, depth does not follow the flow noise level; it remains a clean FP32 structural condition. Nevertheless, features that jointly depend on depth, the noisy video state, and the current flow time are recomputed at every model evaluation.

Under sequential, gap-free traversal and correct source reads, three properties follow (Appendix A). First,

$$C_i = C[a_i : b_i], \tag{12}$$

so overlapping depth positions retain exactly the same canonical latent values and dtype across windows. Second, strictly increasing right endpoints produce disjoint first-read intervals with a total temporal extent of F, independent of the number of flow evaluations K. Therefore, each global depth position is introduced into the rolling state exactly once. Third, the active depth rows satisfy

$$\boxed{M_{\text{depth}} \leq 4L_{\max}S \times 96 \text{ bytes.}} \tag{13}$$

For the default $L_{\max} = 102$, this yields an upper bound of 19.050 MiB at 960×544 resolution and 9.413 MiB at 672×384. The bound covers only the active canonical depth rows and excludes transient copies, fused conditioning features, model activations, and storage owned by the underlying geometry source. Relative to direct slicing of the same canonically encoded source, rolling depth preserves the conditioning tensor exactly. Its role is therefore to organize the active state and avoid redundant access rather than to alter the model output.

3.5 Joint Generation Algorithm

The logical state of window i is

$$\mathcal{S}_i = (\Omega_i, H_i, C_i, I_i, \epsilon_i, c_{\text{g}}) \,. \tag{14}$$

A single global temporal plan determines the indices of video history, depth state, and noise, ensuring that all three refer to the same latent-time positions. Algorithm 1 combines rolling depth, noise-aligned history preservation, and boundary-image updates into a unified continuous generation procedure.

4 Experiments

4.1 Experimental Setup

Data and implementation. The primary set contains 100 thirty-second clips, with 25 each from driving, human activity, simulation, and synthetic scenes. Video Depth Anything (VDA) ViT-L FP32 estimates [Chen et al., 2025] are prepared at 832×480, 16 fps, normalized over each trajectory, and encoded for 960×544 generation. GT follows source timestamps at 24 fps and the same canvas transform for fidelity evaluation; real-time throughput is evaluated against a 14-FPS target. Additional studies cover short-video quality, 24 sources under nine paired conditions, 25 native-depth inputs, 30 long videos, and three PyBullet conditions. Appendix C records source identities, splits, and configured-manifest sensitivity analyses.

The generator combines the released MiniMax-H3 ControlNet-Union for native depth conditioning with the VDN-H3 8-step model obtained by DMD distillation. The Union branch connects to five of 50 H3 blocks;

Algorithm 1 UniWorld continuous generation

Input: I_{ref}, p, depth source $\mathcal{D}$, windows $\{\Omega_i\}$, global noise ϵ, grid $\{t_k\}_{k=0}^{K}$
$c_{\text{g}} \leftarrow E_{\text{Qwen}}(I_{\text{ref}}, p)$; $I_0 \leftarrow I_{\text{ref}}$; $Z \leftarrow [\,]$; depth state $\leftarrow [\,]$
for each $\Omega_i = [a_i, b_i)$ **do**
  $C_i \leftarrow \text{RollDepth}(\mathcal{D}, \Omega_i)$
  $H_i \leftarrow Z[a_i : a_i + h_i]$; $\epsilon_i \leftarrow \epsilon[a_i : b_i]$
  $\mathcal{C}_i \leftarrow \big(C_i, \text{EncodeBoundary}(I_i), c_{\text{g}}\big)$
  $z_i \leftarrow \mathcal{P}_{i,t_0}(\epsilon_i)$
  for $k = 0, \ldots, K-1$ **do**
    $v_i \leftarrow v_\theta(z_i, t_k; \mathcal{C}_i)$
    $z_i \leftarrow \mathcal{P}_{i,t_{k+1}}(z_i + (t_{k+1} - t_k) v_i)$
  end for
  $Z \leftarrow \text{concat}\big(Z, z_i[h_i : L_i]\big)$; depth state $\leftarrow C_i$
  if another window remains: $I_{i+1} \leftarrow \text{DecodeAtNextStart}(Z)$
end for
Decode Z under the global VAE phase and trim alignment padding to the requested delivery length.

Table 1 VBench quality (×100, ↑); best in bold.

Configuration	Quality Score	Subject cons.	Background cons.	Temporal flickering	Motion smoothness	Dynamic degree	Aesthetic quality	Imaging quality
H3 50-step	79.87	**87.46**	92.94	**96.72**	98.47	83.50	52.60	62.28
LongVie2	78.54	86.90	92.86	96.46	97.83	85.00	50.50	58.64
LongVie AR2	78.62	86.28	92.19	96.63	97.84	**86.00**	50.06	60.24
H3/VDN DMD8	**80.54**	87.20	**93.10**	96.54	**98.68**	85.50	**54.57**	**63.51**

inference uses eight evaluations per window, shifts 12/3, and guidance 1. A 30-second output spans three windows and 720 RGB frames. Each worker uses eight H100 GPUs with FSDP/Ulysses, with no additional joint training of the released components.

Metrics and statistics. We report video-level RGB PSNR, SSIM [Wang et al., 2004], and LPIPS [Zhang et al., 2018]. Geometry uses VDA rank correlation, depth-edge F1, and NMAE with calibration fixed after five seconds; native comparisons share inverse-metre GT and original valid masks. Temporal error measures disagreement with GT frame changes at fixed seams. Intervals use 10,000 category-stratified source-group bootstrap replicates, one fixed seed per source, and marginal 95% intervals. Appendix C specifies sampling, calibration, and aggregation.

4.2 Video Quality Comparison

We compare the H3/VDN DMD8 generator used by UniWorld with H3 50-step, LongVie2, and LongVie AR2 on the same 100 sources. Table 1 averages 5- and 14-second outputs equally (200 videos per configuration), using VBench's seven quality dimensions and officially weighted Quality Score [Huang et al., 2024]. DMD8 achieves the highest Quality Score, background consistency, motion smoothness, aesthetic quality, and imaging quality, exceeding H3 50-step by 0.67 Quality Score points. Appendix C.8 gives native configurations, duration-level results, and paired intervals.

4.3 Multi-Scenario Continuous Generation

One checkpoint completes all 100 clips and 300 windows. Table 2 reports source fidelity and depth consistency across the four categories; overall PSNR is 13.3973 dB (95% CI [12.9211, 13.8794]). Figure 4 shows outputs from the first frozen ablation identity in each category. Appendix C provides full intervals and continuation-only measurements.

Table 2 Thirty-second fidelity and depth consistency, 960×544, $W/O = 345/39$, 8 NFE. Rank and Edge denote depth Spearman and edge F1; NMAE fixes calibration after five seconds. The final row excludes configured-manifest overlap.

Set	n	PSNR↑	SSIM↑	LPIPS↓	Rank↑	Edge↑	NMAE↓
Driving	25	13.4429	0.5109	0.4727	0.9042	0.7027	0.0554
Human activity	25	14.9858	0.4799	0.3883	0.9299	0.7509	0.0523
Simulation	25	12.4034	0.3757	0.5914	0.7665	0.5223	0.2524
Synthetic	25	12.7571	0.3612	0.5475	0.9117	0.5655	0.0877
All	100	13.3973	0.4319	0.5000	0.8781	0.6354	0.1120
No manifest overlap	90	13.1587	0.4357	0.5127	0.8702	0.6212	0.1181



Figure 4 One model across four scene categories. Actual reference, GT, and generated frames at the first continuation boundary and final frame. Rows use the first frozen ablation identity per category; selection is independent of quality scores.

Native depth. The same model also completes all 25 PointOdyssey/TartanAir inputs with native metric depth by changing only the source adapter. Appendix C.2 and Table S3 give the matched comparison against common native GT, paired intervals, and valid-support rules.

4.4 Mechanism Analysis and Ablations

State consistency in real model runs. Full-method replay, rolling/direct reads, and aligned depth re-encoding match exactly in all 24 cases; all three online/cached pairs also match. Appendix C.7 and Table S9 retain the complete state checks, including committed history under a control branch.

Fixed-window comparisons. We select six distinct source groups per category by a fixed deterministic ordering. All nine conditions share the exact first-window output, suffix noise, reference/text, window geometry, commitment boundaries, schedule, and VAE. Partial-history variants constrain the final 6 or 2 of the 12 overlap latents, leaving the discarded overlap unchanged. Table 3 reports each alternative minus full per-step projection. Full projection reduces seam LPIPS by 0.00195 relative to boundary RGB only in this paired sample. Continuation PSNR and seam temporal-error intervals include zero, and fresh history noise gives slightly lower seam LPIPS. The endpoint-only control matches boundary-only output exactly: both have unconstrained history during model evaluations and discard the predicted overlap. This isolates per-step history participation from final output preservation. Appendix C.6 provides continuation, geometry, and

Table 3 Paired effects, 24 sources per arm; brackets are marginal 95% bootstrap intervals. PSNR covers [345, 720); LPIPS and temporal error cover fixed seams. Positive LPIPS/error differences favor the full method.

Arm minus full	ΔPSNR (dB)↑	ΔLPIPS↓	ΔE_{temp} ↓
Boundary RGB only	0.00637 [−0.01467, 0.02591]	0.00195 [0.00033, 0.00362]	−0.00047 [−0.00140, 0.00039]
Endpoint only	0.00637 [−0.01467, 0.02591]	0.00195 [0.00033, 0.00362]	−0.00047 [−0.00140, 0.00039]
Fresh history noise	−0.01337 [−0.08561, 0.05934]	−0.00198 [−0.00344, −0.00045]	0.00021 [−0.00054, 0.00103]
Last 6 history latents	0.00623 [−0.00484, 0.01751]	0.00030 [−0.00010, 0.00072]	0.00002 [−0.00022, 0.00029]
Last 2 history latents	0.00917 [−0.00888, 0.02600]	0.00113 [−0.00016, 0.00261]	−0.00023 [−0.00056, 0.00008]
Direct global slice	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]
Window normalization	−0.00897 [−0.04902, 0.03094]	0.00063 [−0.00076, 0.00200]	0.00085 [−0.00014, 0.00210]
Aligned re-encoding	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]

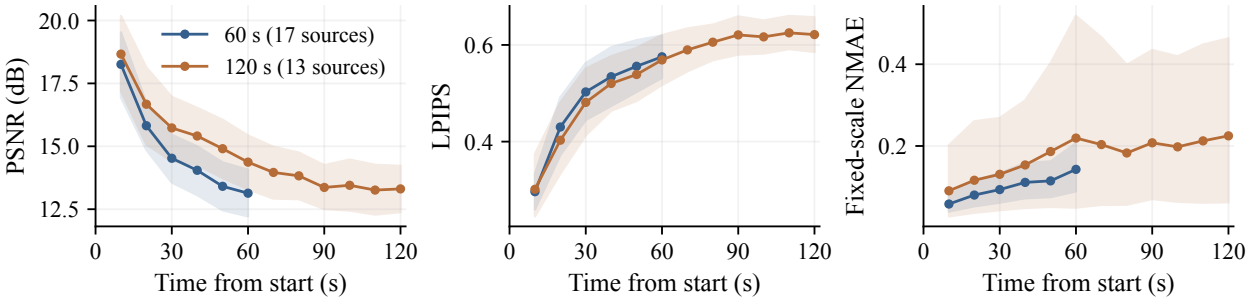


Figure 5 Continuous generation over 60–120 seconds. Ten-second aggregates with category-stratified source-group 95% bootstrap bands. The duration groups use different sources; the 120-second group is a training-split stress test.

manifest-exclusion results under the same protocol.

4.5 Runtime and Resource Usage

DiT forward throughput. Four fixed sources at each resolution receive one full-window warmup and three hot repeats: 32 requests including warmups, 12 hot requests per canvas. We time actual H3/VDN and Union forwards, including FSDP/Ulysses communication, with CUDA synchronization and per-step maxima over eight ranks. The devices are isolated from other experiment jobs; final latents match the corresponding complete-generation runs.

We evaluate real-time throughput against a 14-FPS target. Full-continuation windows produce 14.472 and 21.361 new FPS at 960×544 and 672×384, respectively, meeting the target at both resolutions. These timings cover synchronized H3/VDN and Union forwards with FSDP/Ulysses communication on eight H100 GPUs; Appendix D reports the complete window profiles and whole-trajectory throughput.

Depth-state resources. Active depth rows occupy 19.050 MiB at 960×544 and 9.413 MiB at 672×384, independently of trajectory length. Rolling reads 217/427/852 positions for 30/60/120-second outputs, versus 241/475/960 for direct slicing, while preserving identical output. This provides a bounded geometric working state and avoids repeated source requests. Consumer copies, carrier, hints, model storage, and growing global caches are accounted for separately, together with measured reader-switch costs, in Appendix D.2.

4.6 Long-Horizon Behaviour and Applications

The same window interface advances through 17 sixty-second videos from existing validation/test splits and thirteen 120-second training-split stress tests. Their full-video PSNR/LPIPS are 14.4542/0.49032 and 14.4277/0.54645. Figure 5 characterizes fidelity and fixed-calibration geometric agreement over successive ten-second blocks; each duration uses its own source set and source-group confidence bands.

Three procedural PyBullet conditions [Coumans and Bai, 2016–2021] exercise warm/cool reference appearance, a camera revisit at frame 480, and a control branch activated at the second-window request, affecting frame 357 onward. On-demand and cached depth give bitwise-identical final latents in all three pairs. The control branch preserves the first window exactly and changes the new suffix (latent mean absolute

difference 0.039076). Warm/cool references share identical geometry while producing different appearance. The first request reads 357 geometry frames, and RGB is delivered incrementally with the canonical decoder phase. Appendix D.4 reports the appearance, revisit, and response measurements for this window-level engine interface.

5 Conclusion

We presented UniWorld, a unified framework for continuous visual world simulation from depth trajectories and reference appearance. Its joint state contract preserves committed video latents, retains matching geometric controls, and reads new depth positions incrementally with bounded active state. One shared generator connects estimated depth, native simulation data, and an on-demand engine; exact replay and cached/online equivalence verify this connection in real model runs. Multi-scenario generation, fixed-window interventions, and long-horizon trajectories characterize the resulting visual simulation, while stage-specific measurements make its computation and data-access costs explicit. The measured steady-state DiT path achieves real-time throughput while maintaining bounded active state for continuous generation.

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A Proofs and State Invariants

A.1 History Projection

We regard the binary mask M_i as broadcast over channel and spatial dimensions. The saved history H_i and global noise slice ϵ_i are fixed within a window. All values are finite; statements below are exact in real arithmetic. Define $Q_{i,t} = tH_i + (1-t)\epsilon_i$.

Proposition S1 (Idempotence and per-step history constraint)**.** *For fixed i, t, $\mathcal{P}_{i,t}$ is idempotent. The projected sampler satisfies $M_i \odot z_i^{(k)} = M_i \odot Q_{i,t_k}$ for every $k = 0, \ldots, K$, and its terminal history is H_i.*

Proof. Since $M_i^2 = M_i$ and $M_i \odot (1 - M_i) = 0$,

$$\mathcal{P}_{i,t}(\mathcal{P}_{i,t}(z)) = M_i \odot Q_{i,t} + (1 - M_i) \odot [M_i \odot Q_{i,t} + (1 - M_i) \odot z] \tag{S1}$$

$$= M_i \odot Q_{i,t} + (1 - M_i) \odot z = \mathcal{P}_{i,t}(z). \tag{S2}$$

Initialization projects at t_0, establishing the constraint for $k = 0$. Each subsequent update ends by applying $\mathcal{P}_{i,t_{k+1}}$, which replaces every masked entry by the corresponding entry of $Q_{i,t_{k+1}}$, regardless of the predicted velocity. This proves the induction step. Finally $t_K = 1$, so $Q_{i,t_K} = H_i$ on the prefix. □

Equation 9 separately preserves the stored global prefix by concatenating only new positions. Thus global committed history does not depend on recovering the sampled prefix at the end. Per-step projection serves the distinct purpose of placing that history in every model evaluation. At $h_i = 0$, $M_i = 0$ and the projection is the identity. We require $h_i < L_i$, so each continuation adds content.

Finite precision. The implementation owns FP32 copies of history and noise and writes $tH_i + (1-t)\epsilon_i$ directly into prefix rows. Repeated projections at the same numerical t use the same operands. At $t = 1$, the finite prefix values are recovered without a dtype conversion; the stored committed prefix is never rewritten. Claims of bitwise equality refer to the actual tensor checks under this implementation. They do not imply exact real-valued flow integration, equality under arbitrary precision changes, or equality of RGB decoded with different temporal contexts. The operation changes a sampling boundary condition, not the learned velocity field, and does not guarantee a globally coherent scene representation.

A.2 Depth Overlap and Incremental Reads

Let a canonical depth sequence C have length F, and let the windows satisfy $a_0 = 0$, $b_{J-1} = F$, $a_i < a_{i+1} \leq b_i < b_{i+1}$, and $0 \leq h_{i+1} < L_{i+1}$. Each read returns the requested canonical values in the same dtype and order.

Proposition S2 (Canonical-condition equivalence)**.** *If $C_0 = C[a_0 : b_0]$, equation 10 gives $C_i = C[a_i : b_i]$ for every window. In particular, $C_i[0 : h_i] = C_{i-1}[a_i - a_{i-1} : L_{i-1}]$.*

Proof. The base case is assumed. If the statement holds for $i - 1$, its retained suffix is exactly $C[a_i : b_{i-1}]$. The new read is $C[b_{i-1} : b_i]$. These intervals meet at b_{i-1} without a gap or duplicated element. Their concatenation is $C[a_i : b_i]$, proving the induction step. Identical-value copying preserves the overlapping FP32 bit patterns. □

Proposition S3 (Read count and active storage)**.** *During one successful sequential traversal, first reads cover F temporal positions, and one active depth state uses at most $4L_{\max}S \times 96$ bytes.*

Proof. The intervals are $[0, b_0), [b_0, b_1), \ldots, [b_{J-2}, F)$. They are disjoint and partition $[0, F)$, giving $b_0 + \sum_{i=1}^{J-1}(b_i - b_{i-1}) = F$ positions. No model evaluation reads an additional interval because C_i is fixed within the window. The active tensor contains at most $L_{\max}S$ rows, each with 96 FP32 elements of four bytes. □

First-read counts concern logical source positions, not operating-system disk traffic or GPU copies. A direct source slice can be a view; it need not perform physical I/O. Concatenation, defensive return copies, and carrier packing still cost time and memory. A failed read may be retried, and restarting a request may reread its source. The stated count applies to successful traversal and same-window cache reuse, without such restarts. Likewise, transient old-plus-new buffers can exceed the size of a single active state.

B Conditioning and Implementation

B.1 Model Components

The backbone is MiniMax-H3 with VDN hybrid video attention [MiniMax, 2026, Xi et al., 2026]. The 50-block backbone retains bidirectional within-window processing. The five-block dense Union branch comes from MiniMax-H3-Fun-Controlnet-Union in VideoX-Fun [aigc-apps, 2026]; its parameters, Q/K/V projections, and activations remain separate. The original Qwen3-VL-32B encoder [Bai et al., 2025] provides layer-50 hidden states of width 5,120 for image and text conditioning. The existing video VAE supplies both the RGB and depth encodings.

The shared generator combines published VDN weights with the existing H3 and Union components, without additional local joint training or a new distillation objective. Appendix C describes evaluation-source selection and the scope of the local training-overlap analysis.

Experiments use PyTorch 2.11 with CUDA 12.8 on workers equipped with eight 80-GB NVIDIA H100 GPUs. Metric and simulator versions are specified with their evaluation protocols.

B.2 Independent First-Frame Conditions

The original Qwen representation stays fixed across windows. The single-frame VAE path uses posterior sampling with encoding seed 42, FP16 rounding, then FP32 latent normalization. Denote the resulting row tensor by R_i. The fixed first-frame input is

$$c_i^{\text{ref}} = 0.999R_i + 0.001\epsilon^{\text{ref}}, \qquad t_i^{\text{ref}} = \max(t, 0.999). \tag{S3}$$

The noise realization is fixed for the request. The input rows remain fixed within a window even though their model activations change. This differs from target-prefix projection, whose values explicitly follow t from 0 to 1. Casting 0.999 to BF16 first would round it to 1.0; the prescribed condition mixture and history projection therefore use FP32 arithmetic. The model retains its native modulation conventions. We do not refresh the Qwen embedding from each generated boundary or use GT boundaries during inference.

B.3 Depth Rows and Union Packing

Depth uses posterior *mode*, followed by FP32 normalization and $(1, 2, 2)$ patchification. Unlike equation S3, depth is clean conditioning without the first-frame mixture. The Union input has 49 pre-patch channels: 24 for depth, one for a visibility/mask channel, and 24 for inpainting. Spatial patchification gives $49 \times 4 = 196$ carrier columns. The 96 depth columns occupy the target-video rows; the remaining 100 columns are zero for this path. A first-frame placeholder is included in the carrier layout.

Target rows are selected by explicit indices. Qwen vision tokens, independent VAE condition rows, and target video rows can share modality tags, so a tag test alone cannot identify the target. Zero padding must also be combined with explicit row selection to limit the effect of a learned projection bias. The depth tensor and control carrier are reused within a window, but Union hints depend on the current noisy target and time and are recomputed at every step. There is no cross-window backbone K/V cache introduced by the two state rules.

Table S1 Analytical storage for FP32 depth rows. Carrier storage, when listed, includes one first-frame placeholder; transient copies and model state are excluded.

Canvas	S	102-latent depth	12-latent overlap	196-column carrier
672×384	252	9.413 MiB	1.107 MiB	19.407 MiB
960×544	510	19.050 MiB	2.241 MiB	39.276 MiB

B.4 Sampling and State Lifecycle

Each video window uses eight actual velocity evaluations with video shift 12; the audio shift is 3 and guidance scale is 1. A scheduler interface that counts grid points may expose nine points for eight updates. Global video and audio noise and first-frame noise are prepared deterministically and sliced consistently. Projection runs once before the first evaluation and once after each update, so the next model input already satisfies the constraint. The first-window mask is empty.

Rolling depth validates ordered window indices, global starts, VAE/audio phase, overlap, shape, FP32 dtype, and finite newly read values. Valid data are committed atomically to the logical cache; returning a copy prevents callers from modifying retained controls through shared storage. Retrying the current window reuses that state.

C Evaluation Protocols and Additional Comparisons

C.1 Datasets and Source Selection

The primary set comprises the sources in Table S2. Named collections include DanceTrack [Sun et al., 2022], BEHAVE [Bhatnagar et al., 2022], PointOdyssey [Zheng et al., 2023], TartanAir [Wang et al., 2020], and MIND [Ye et al., 2026]. We use PointOdyssey v1.2 and TartanAir V2 as part of this common evaluation set rather than their official benchmark protocols. The human-activity category also includes eleven local original clips.

Table S2 Primary 30-second sources. Every category initially uses estimated VDA depth. Native-depth inputs are a paired intervention on the same 25 simulation sources.

Category	Clips	Collection
Driving	25	Nexar
Human activity	25	DanceTrack (10), BEHAVE (4), local originals (11)
Simulation	25	PointOdyssey v1.2 (15), TartanAir V2 (10)
Synthetic	25	MIND (25)

The 100 primary clips contain 96 distinct source groups. Ten human-activity clips also appear in the configured 10,000-sample local training set. The evaluated generator reuses pretrained components without local joint training; exclusion of these clips is therefore reported as a sensitivity analysis. This overlap check covers the local data collection and does not characterize the original pretraining data of H3, VDN, or Union.

The 24 ablation sources comprise six distinct source groups per category, selected by a fixed deterministic ordering independent of quality. Every source uses one deterministic seed. The 17 sixty-second sources use existing validation/test splits; all 13 120-second sources belong to an existing training split and are reported as stress tests.

C.2 Depth Representation and Ground-Truth Alignment

Estimated depth uses VDA ViT-L FP32 over the full trajectory at 832×480, 16 fps. Normalized depth is temporally resampled and encoded on the generation canvas. RGB GT is resampled at 24 fps using source timestamps, with an 832×480 cover resize/crop followed by 960×544 Lanczos resizing for primary comparisons. This 24-FPS stream is retained for fidelity evaluation; real-time throughput is evaluated against a 14-FPS target. Generated videos are evaluated at their delivered resolution and duration, including the generated first frame. Padding needed for $17n + 5$ geometry is never included in delivered-frame counts.

Verified numerical direction. VDA raw output is inverse-depth oriented, with larger values near the camera. The normalization helper reverses that ordering, making the actual condition *far bright*. For metric depth $z > 0$, native preparation uses $u = 1/z$, obtains trajectory-wide q_5, q_{95} from originally valid inverse-depth

pixels, and applies

$$\bar{D} = \frac{q_{95} - \mathrm{clip}(u, q_5, q_{95})}{q_{95} - q_5}. \tag{S4}$$

This matches the checkpoint's existing numeric convention; it does not change the VDA baseline. Native TartanAir uses packed float32 metres; PointOdyssey depth uses $z = 1000\, d_{16}/65535$ metres. Invalid pixels are filled temporally for generation only, and original masks accompany the metrics. All 25 native comparisons retain identical prompt embeddings, text tags, first-frame rows, reference, noise identity, and model; only the depth condition changes.

Table S3 Estimated versus native depth on 25 matched sources; geometry uses common native GT.

Input	PSNR↑	SSIM↑	LPIPS↓	Rank↑	Edge↑	NMAE↓
Estimated VDA	12.4034	0.3757	0.5914	0.7818	0.4951	0.2783
Native metric depth	11.6948	0.3527	0.6208	0.7778	0.4798	0.2722

For completeness, the native-minus-estimated paired differences are −0.70859 dB PSNR (95% CI $[-1.31984, -0.04591]$), +0.02943 LPIPS ($[0.00569, 0.05236]$), −0.00408 native-GT rank ($[-0.03886, 0.04161]$), and −0.00612 native-GT NMAE ($[-0.05175, 0.03045]$). These use the same 25 sources and 21 source groups, with common native GT for both geometry evaluations.

On-demand source. PyBullet 3.2.7 TinyRenderer converts its z-buffer to axial metre depth using near/far planes 0.1/25 m. A fixed, predeclared 1.5/12 m mapping supplies far-bright conditions:

$$\bar{D}_{\mathrm{engine}}(z) = \mathrm{clip}\left(\frac{1/1.5 - 1/z}{1/1.5 - 1/12}, 0, 1\right). \tag{S5}$$

It is quantized to uint8 and encoded in canonical blocks. This adapter uses fixed metric bounds and does not inspect future trajectory percentiles. Warm and cool appearances have bitwise-identical depth, and the control branch shares the initial source interval exactly.

C.3 Fidelity, Perceptual, and Temporal Metrics

For video s, RGB values in $[0, 1]$, and frame set $\mathcal{T}$, the primary PSNR is

$$\mathrm{MSE}_s(\mathcal{T}) = \frac{1}{3H_{\mathrm{px}}W_{\mathrm{px}}|\mathcal{T}|} \sum_{j \in \mathcal{T}} \sum_{x,y,c} (\hat{V}_{s,j,x,y,c} - V_{s,j,x,y,c})^2, \tag{S6}$$

$$\mathrm{PSNR}_s(\mathcal{T}) = -10 \log_{10} \mathrm{MSE}_s(\mathcal{T}). \tag{S7}$$

Decoded eight-bit RGB is converted to FP64 for this calculation. We average video-level dB, not framewise dB or the dB of pooled cross-video MSE. SSIM uses FFmpeg's default 8×8 blocks over all RGB channels, averaged across frames then videos. Full, first-frame-excluded, and common continuation sets are $[0, 720)$, $[1, 720)$, and $[345, 720)$.

LPIPS uses the official v0.1 AlexNet model (package 0.1.4), RGB scaled to $[-1, 1]$, at the original inference canvas. Its deterministic sample contains every twelfth frame at 24 fps plus all frames in the fixed seam bands. The full LPIPS mean therefore includes this seam-augmented sample; seam LPIPS is reported separately. Temporal error uses every 24-fps consecutive-frame transition:

$$E_{\mathrm{temp}}(j) = \mathrm{mean}_{x,y,c} \left| (\hat{V}_j - \hat{V}_{j-1}) - (V_j - V_{j-1}) \right|. \tag{S8}$$

Prediction and GT motion magnitudes are recorded separately. Small frame-to-frame change alone does not establish correct motion. Fixed 12-frame neighborhoods are $[339, 351)$ and $[645, 657)$; all ablations use identical absolute indices.

Across the primary set, SSIM is 0.431918, LPIPS is 0.49997, and depth rank correlation is 0.87809. Excluding frame zero gives 13.3915 dB PSNR; the common continuation $[345, 720)$ gives 12.6700 dB PSNR and 0.54241 LPIPS.

C.4 Geometric Metrics and Valid Support

Generated RGB is sampled uniformly at 2 fps, resized to 832×480, and passed to VDA ViT-L FP32 with internal input size 518. Prediction and target depth are area-resized to 208×120 for metrics. Rank is Spearman correlation. Edge F1 thresholds each depth-gradient magnitude at its valid-region 90th percentile and permits a one-pixel match radius. A nonnegative affine scale and offset are fitted once over valid pixels in the first five seconds; later frames never refit. NMAE divides fixed-calibration absolute error by the calibration target's $p_{95} - p_5$.

Primary geometry compares generated-video VDA output against the VDA conditioning trajectory and measures estimator self-consistency. In native comparisons, both generated outputs use the *same native inverse-metre target*; scores against different conditioning sources cannot establish source-to-source geometric gains. This native target removes dependence on VDA for target geometry, while the generated-side estimate still has estimator bias.

A downsampled native-depth pixel is valid only if every original pixel contributing to its area average is valid. Edge evaluation additionally requires a fully valid 3×3 Sobel neighborhood. Filled values never participate in calibration or evaluation. Frames without valid support or calibration contribute no value to the corresponding metric; aggregate means use the remaining valid observations and retain their sample counts.

C.5 Cluster Bootstrap and Aggregate Intervals

Bootstrap uses seed 20260924 and 10,000 replicates. We sample source groups with replacement within each category, preserving all videos and their weights in each selected group. For ablations, resampling uses paired same-source differences. This estimates variation across sources under a fixed seed per source, not seed-to-seed generation variability. Intervals are marginal 95% percentiles, with no family-wise adjustment across the exploratory arms and metrics.

Table S4 Primary source-fidelity means with category-stratified source-group 95% bootstrap intervals. The 90-clip subset excludes overlap with the configured local training set.

Set	PSNR (dB)↑	SSIM↑	LPIPS↓
Driving	13.4429 [12.4759, 14.4184]	0.5109 [0.4682, 0.5516]	0.4727 [0.4283, 0.5158]
Human activity	14.9858 [13.9759, 15.9918]	0.4799 [0.4163, 0.5395]	0.3883 [0.3499, 0.4316]
Simulation	12.4034 [11.2682, 13.5804]	0.3757 [0.3093, 0.4445]	0.5914 [0.5529, 0.6262]
Synthetic	12.7571 [12.0658, 13.4274]	0.3612 [0.3073, 0.4178]	0.5475 [0.5238, 0.5701]
All	13.3973 [12.9211, 13.8794]	0.4319 [0.4037, 0.4601]	0.5000 [0.4813, 0.5185]
No manifest overlap	13.1587 [12.6365, 13.6800]	0.4357 [0.4063, 0.4653]	0.5127 [0.4926, 0.5329]

C.6 Paired Mechanism Protocol and Additional Results

All 216 paired outputs use the same first-window video/audio latents for their source. Non-depth inputs, suffix noise, window size, discarded overlap, commitment points, 8 NFE, video/audio schedule 12/3, and VAE precision remain fixed. The depth interventions alter only the intended continuation conditions. Table S5 specifies the interventions.

At the endpoint, restoring history cannot alter the suffix already generated. Boundary-only and endpoint-only variants therefore produce identical committed outputs, as confirmed in 24/24 cases. Direct slicing and rolling retain identical conditions and outputs, also in 24/24 cases. Independent phase-aligned re-encoding is exact because the H3 encoder uses independent 17-frame chunks; these cases do not demonstrate encoding-context drift. Comparisons that vary both overlap and window size are reported separately in Appendix D.5.

The seam comparison measures history participation near fixed output boundaries, while the continuation table measures its effect over the full suffix. The boundary-only seam LPIPS difference is 0.00195 on all 24 sources and 0.00169 on the 22-source sensitivity subset; the latter interval includes zero. Its continuation temporal-error difference is -0.000324 ($[-0.000449, -0.000195]$). Reporting both temporal scopes separates

Table S5 Nine completed conditions, each applied to the same 24 sources. Geometry and commitment boundaries remain fixed.

Condition	Intervention
Full	Project all 12 history latents at every sampling step; canonical rolling depth.
Boundary RGB only	Retain independent boundary-image conditioning; no target-history projection.
Endpoint only	Project the full overlap only at $t = 1$, after all model evaluations.
Fresh history noise	Redraw overlap noise once, then keep it fixed across steps; suffix noise is unchanged.
Last 6 / last 2 latents	Project only $[h_i - q, h_i)$ with $q = 6$ or 2, still discarding all h_i overlap latents.
Direct global slice	Read canonical $C[a_i : b_i]$ independently at each advance.
Window normalization	Recompute percentiles for continuation windows with canonical global encoding context.
Aligned re-encoding	Keep globally normalized pixels; encode each phase-aligned window independently with its padded tail.

Table S6 Additional continuation metrics on $[345, 720)$, alternative minus full, $n = 24$ per row. Brackets give marginal 95% source-group bootstrap intervals.

Arm minus full	ΔSSIM↑	ΔLPIPS↓	ΔE_{temp} ↓
Boundary RGB only	0.00018 [−0.00052, 0.00083]	0.00157 [0.00036, 0.00288]	−0.00032 [−0.00045, −0.00020]
Endpoint only	0.00018 [−0.00052, 0.00083]	0.00157 [0.00036, 0.00288]	−0.00032 [−0.00045, −0.00020]
Fresh history noise	−0.00357 [−0.00861, 0.00032]	−0.00118 [−0.00266, 0.00045]	0.00000 [−0.00042, 0.00039]
Last 6 history latents	0.00045 [0.00001, 0.00083]	0.00026 [−0.00009, 0.00064]	−0.00001 [−0.00009, 0.00008]
Last 2 history latents	0.00049 [−0.00011, 0.00103]	0.00099 [0.00001, 0.00208]	−0.00030 [−0.00044, −0.00017]
Direct global slice	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]
Window normalization	−0.00050 [−0.00278, 0.00222]	0.00052 [−0.00073, 0.00182]	0.00048 [−0.00002, 0.00110]
Aligned re-encoding	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]

Table S7 Full-video VDA geometry effects, alternative minus full, $n = 24$ per row. These measure consistency with VDA input, not independent metric geometry. Intervals are unadjusted across metrics and arms.

Arm minus full	ΔRank↑	ΔEdge↑	ΔNMAE↓
Boundary RGB only	−0.00091 [−0.00158, −0.00031]	−0.00078 [−0.00181, 0.00028]	0.00032 [−0.00023, 0.00084]
Endpoint only	−0.00091 [−0.00158, −0.00031]	−0.00078 [−0.00181, 0.00028]	0.00032 [−0.00023, 0.00084]
Fresh history noise	−0.00103 [−0.00361, 0.00093]	0.00011 [−0.00281, 0.00295]	−0.00042 [−0.00164, 0.00086]
Last 6 history latents	−0.00021 [−0.00138, 0.00095]	0.00024 [−0.00060, 0.00118]	−0.00018 [−0.00081, 0.00033]
Last 2 history latents	−0.00084 [−0.00232, 0.00016]	−0.00083 [−0.00153, −0.00014]	−0.00011 [−0.00071, 0.00047]
Direct global slice	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]
Window normalization	0.00175 [−0.00015, 0.00380]	−0.00046 [−0.00170, 0.00082]	−0.00158 [−0.00564, 0.00126]
Aligned re-encoding	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]	0.00000 [0.00000, 0.00000]

Table S8 Selected paired effects after excluding overlap with the configured local training set ($n = 22$). PSNR is continuation fidelity; LPIPS and temporal error use fixed seams.

Arm minus full	n	ΔPSNR↑	ΔLPIPS↓	ΔE_{temp} ↓
Boundary RGB only	22	0.00462 [−0.01798, 0.02533]	0.00169 [−0.00001, 0.00331]	−0.00051 [−0.00152, 0.00044]
Fresh history noise	22	−0.01197 [−0.08981, 0.06875]	−0.00170 [−0.00328, −0.00012]	0.00018 [−0.00061, 0.00107]
Window normalization	22	−0.01149 [−0.05585, 0.03268]	0.00075 [−0.00075, 0.00229]	0.00086 [−0.00022, 0.00224]

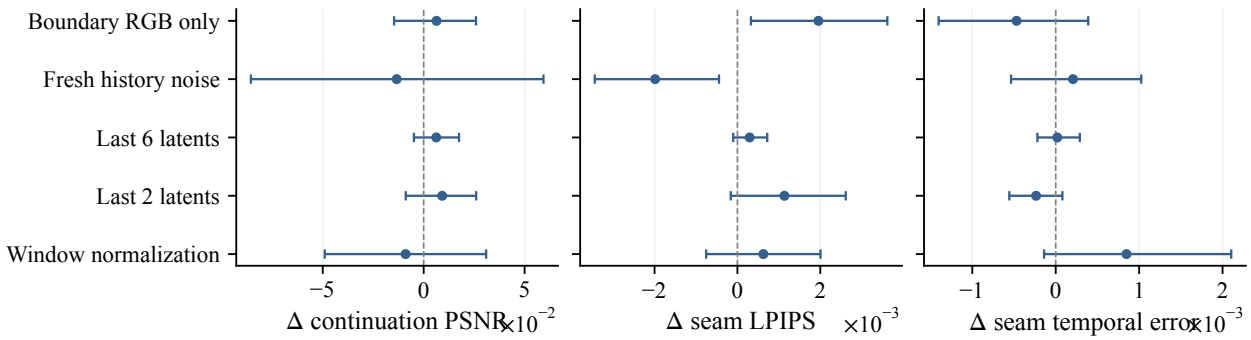


Figure S1 Paired effect sizes and marginal 95% intervals for non-equivalent interventions, each minus full history projection. Endpoint-only is identical to boundary-only; direct slicing and aligned re-encoding are exact zero-effect controls.

local seam behavior from the complete generated interval. State preservation is established by the exact checks independently of these quality measurements.

C.7 State Consistency Checks

All 24 full-method replays reproduce their primary outputs exactly. Full and native-depth windows execute eight model evaluations; each alternative arm reuses its source's first-window output and executes eight evaluations in each continuation window. Constrained history rows have zero maximum absolute error. Local boundary extraction matches full decoding at window lengths of 92, 97, and 102 latents. Rolling and direct depth access also produce bitwise-equal video and audio latents in two 60-second cases. The quality evaluation concerns the visual outputs.

Table S9 Temporal state consistency in completed model runs. Equality is checked on latent tensors or complete video outputs.

Property	Result
Reconstruct canonical generation	24/24 full-method outputs exactly match
Incremental geometric reads	24/24 rolling/direct outputs bitwise equal
Canonical encoding phase	24/24 aligned re-encoded outputs bitwise equal
On-demand source equivalence	3/3 online/cached outputs bitwise equal
Committed control-branch history	Initial-window latents bitwise equal

C.8 VBench Quality Comparison: Protocol and Paired Results

Table 1 evaluates 800 existing videos from four configurations, two nominal durations, and the same 100 sources (25 per category; 96 source groups). The H3/VDN DMD8 row is the shared eight-evaluation generator used by UniWorld. Its 5/14-second outputs occupy one window, characterizing within-window generation; Section 4.4 evaluates the separate continuation mechanisms. The comparison retains each configuration's original weights, sampling, spatial canvas, and temporal rate.

All configurations use reference RGB and depth; LongVie additionally receives SpaTracker trajectories. The H3 pair shares input caches and seed policy, with different generator weights and attention implementations. The DMD8 checkpoint has zero local joint-training updates. H3 uses 124/345 internal frames before trimming. LongVie2 uses 81-frame windows with one-frame overlap, and its 5-second output is the first window of the 14-second run. LongVie AR2 uses a finite 12-latent window with eight sink latents. Preserving the LongVie endpoint gives encoded durations of 5.0625/14.0625 seconds. The results describe these complete configurations; their native resolutions, frame rates, and control inputs are part of the comparison.

Table S10 Generation configurations for VBench.

Configuration	Canvas	FPS	Frames (5/14 s)	Sampling
H3 50-step	960 × 544	24	120/336	50 steps, CFG 1
H3/VDN DMD8	960 × 544	24	120/336	8 NFE, CFG 1
LongVie2	640 × 352	16	81/225	50 steps, CFG 5
LongVie AR2	640 × 352	16	81/225	2 steps per latent, CFG 1

Official evaluation and aggregation. We use VBench 0.1.5 [Huang et al., 2024] with its official custom-video evaluation pipeline. Each metric reads the original video through its official loader. DINO and CLIP use the official frame transforms; MUSIQ uses the longer-side resizing policy; RAFT samples at approximately 8 fps; AMT-S uses original adjacent-frame intervals. Pixel-change flickering scores on these moving scenes are interpreted jointly with dynamic degree, which records detected movement.

For dimension score s_d, official bounds m_d, M_d, and weight w_d, the reported Quality Score is

$$Q = \frac{1}{6.5}\sum_{d=1}^{7} w_d \frac{s_d - m_d}{M_d - m_d}, \qquad w_{\text{dynamic}} = 0.5, \quad w_d = 1 \text{ otherwise.} \tag{S9}$$

Normalized values are retained without clipping. The bounds for subject consistency, background consistency, flickering, and smoothness are respectively $[0.1462, 1]$, $[0.2615, 1]$, $[0.6293, 1]$, and $[0.706, 0.9975]$; the other dimensions use $[0, 1]$. Raw MUSIQ values are divided by 100 once. Equal-video dimension means are aggregated with equation S9, and all table values are multiplied by 100. This is the quality component on our custom source collection; source fidelity and depth agreement are evaluated separately in Tables 2 and S3.

For the combined results in Table 1, we average the two duration scores within each source, then average over the 100 sources with equal weight. With balanced counts, this equals the mean over all 200 videos per configuration and gives each duration half the total weight. Table S11 retains the original duration-level means.

Table S11 VBench quality by duration (×100, ↑); best in bold.

Configuration	Quality Score	Subject cons.	Background cons.	Temporal flickering	Motion smoothness	Dynamic degree	Aesthetic quality	Imaging quality
5 s; n = 100 per configuration								
H3 50-step	80.40	**89.38**	93.78	**96.92**	98.60	84.00	52.77	60.93
LongVie2	79.12	88.64	93.59	96.45	97.83	86.00	50.57	58.83
LongVie AR2	79.04	87.37	92.65	96.64	97.84	**88.00**	50.20	59.92
H3/VDN DMD8	**80.89**	88.93	**93.82**	96.74	**98.75**	86.00	**54.06**	**62.31**
14 s; n = 100 per configuration								
H3 50-step	79.34	**85.55**	92.10	96.53	98.34	83.00	52.42	63.64
LongVie2	77.96	85.17	92.13	96.47	97.82	84.00	50.43	58.45
LongVie AR2	78.19	85.19	91.73	**96.62**	97.84	84.00	49.92	60.56
H3/VDN DMD8	**80.18**	85.48	**92.37**	96.35	**98.60**	**85.00**	**55.08**	**64.71**

Paired uncertainty. Tables S12 and S13 report DMD8 minus each comparator on the same sources. We resample source groups within each category, retain fixed category weights and all model/metric pairings, and evaluate each duration separately. There are 10,000 bootstrap replicates with seed 20260925. Brackets are marginal 95% percentile intervals without multiplicity adjustment. One fixed output per source/configuration contributes to each duration. All six Quality Score intervals are positive; Tables S14 and S15 retain the full category-level means.

Table S12 Paired differences (DMD8 − comparator, ×100); brackets: 95% intervals.

Configuration	Quality Score	Subject cons.	Background cons.	Temporal flickering
5 s: DMD8 minus comparator; $n = 100$ pairs				
	+0.49	−0.45	+0.04	−0.19
H3 50-step	[+0.15, +0.85]	[−0.81, −0.12]	[−0.16, +0.25]	[−0.29, −0.08]
	+1.78	+0.29	+0.23	+0.29
LongVie2	[+1.39, +2.16]	[−0.25, +0.78]	[−0.02, +0.48]	[+0.11, +0.45]
	+1.85	+1.55	+1.17	+0.10
LongVie AR2	[+1.42, +2.26]	[+0.96, +2.09]	[+0.84, +1.50]	[−0.05, +0.25]
14 s: DMD8 minus comparator; $n = 100$ pairs				
	+0.84	−0.07	+0.28	−0.18
H3 50-step	[+0.48, +1.21]	[−0.74, +0.57]	[−0.19, +0.76]	[−0.27, −0.09]
	+2.22	+0.31	+0.24	−0.13
LongVie2	[+1.84, +2.61]	[−0.40, +1.01]	[−0.17, +0.65]	[−0.36, +0.08]
	+1.98	+0.29	+0.64	−0.27
LongVie AR2	[+1.55, +2.44]	[−0.39, +0.95]	[+0.23, +1.05]	[−0.50, −0.05]

Table S13 Remaining paired VBench dimensions; notation as in Table S12.

Configuration	Motion smoothness	Dynamic degree	Aesthetic quality	Imaging quality
5 s: DMD8 minus comparator; $n = 100$ pairs				
	+0.15	+2.00	+1.29	+1.38
H3 50-step	[+0.09, +0.22]	[−1.45, +6.00]	[+0.81, +1.77]	[+0.62, +2.26]
	+0.92	+0.00	+3.49	+3.48
LongVie2	[+0.72, +1.14]	[−3.00, +3.00]	[+2.68, +4.28]	[+2.05, +4.95]
	+0.91	−2.00	+3.86	+2.39
LongVie AR2	[+0.73, +1.12]	[−6.00, +2.00]	[+3.11, +4.59]	[+1.30, +3.51]
14 s: DMD8 minus comparator; $n = 100$ pairs				
	+0.26	+2.00	+2.67	+1.07
H3 50-step	[+0.18, +0.35]	[+0.00, +5.00]	[+1.84, +3.50]	[+0.29, +1.89]
	+0.78	+1.00	+4.65	+6.26
LongVie2	[+0.61, +0.96]	[+0.00, +3.00]	[+3.77, +5.49]	[+4.85, +7.79]
	+0.76	+1.00	+5.16	+4.15
LongVie AR2	[+0.60, +0.94]	[−2.00, +4.00]	[+4.20, +6.09]	[+2.89, +5.46]

Table S14 VBench quality for driving and human activity (×100, ↑); best in bold.

Configuration	Quality Score	Subject cons.	Background cons.	Temporal flickering	Motion smoothness	Dynamic degree	Aesthetic quality	Imaging quality
Driving: 5 s; $n = 25$ per configuration								
H3 50-step	79.34	**93.94**	96.14	**98.39**	99.12	76.00	44.42	52.12
LongVie2	79.29	93.20	96.02	97.72	98.81	**84.00**	43.58	52.53
LongVie AR2	79.02	90.72	94.62	97.74	98.87	**84.00**	45.05	53.87
H3/VDN DMD8	**80.31**	93.91	**96.37**	98.20	**99.16**	80.00	**46.36**	**54.56**
Driving: 14 s; $n = 25$ per configuration								
H3 50-step	78.93	**90.71**	94.76	**98.21**	99.06	80.00	44.00	54.25
LongVie2	78.43	90.61	94.66	97.59	98.71	84.00	44.42	51.69
LongVie AR2	78.12	88.72	93.41	97.67	98.80	80.00	44.43	55.02
H3/VDN DMD8	**80.22**	90.26	**94.92**	97.99	**99.09**	**88.00**	**47.06**	**56.32**
Human activity: 5 s; $n = 25$ per configuration								
H3 50-step	81.11	**93.34**	94.47	98.52	99.25	60.00	54.36	63.83
LongVie2	80.21	93.26	**94.72**	98.03	98.82	64.00	52.67	60.24
LongVie AR2	80.50	92.54	94.12	97.96	98.78	**68.00**	52.34	62.41
H3/VDN DMD8	**81.95**	92.86	94.43	**98.53**	**99.29**	**68.00**	**55.18**	**64.95**
Human activity: 14 s; $n = 25$ per configuration								
H3 50-step	80.45	91.00	92.77	98.23	99.07	60.00	54.70	65.71
LongVie2	79.49	**91.60**	**93.99**	98.13	98.87	60.00	52.89	59.81
LongVie AR2	79.79	91.56	93.53	97.96	98.76	**64.00**	51.69	62.51
H3/VDN DMD8	**80.96**	90.70	93.31	**98.24**	**99.20**	60.00	**56.34**	**66.50**

Table S15 Simulation and synthetic VBench quality; notation as in Table S14.

Configuration	Quality Score	Subject cons.	Background cons.	Temporal flickering	Motion smoothness	Dynamic degree	Aesthetic quality	Imaging quality
Simulation: 5 s; $n = 25$ per configuration								
H3 50-step	78.94	**82.29**	**91.15**	95.71	97.91	**100.00**	51.03	62.68
LongVie2	76.72	81.07	90.00	95.23	96.33	96.00	48.53	62.46
LongVie AR2	77.04	79.94	89.39	**96.00**	96.76	**100.00**	47.74	61.97
H3/VDN DMD8	**79.28**	82.20	91.05	95.70	**98.31**	96.00	**52.97**	**63.83**
Simulation: 14 s; $n = 25$ per configuration								
H3 50-step	77.39	77.65	**89.25**	95.14	97.53	**92.00**	51.76	66.71
LongVie2	75.45	76.92	88.63	95.35	96.42	**92.00**	48.18	62.62
LongVie AR2	76.08	77.65	88.67	**95.95**	96.81	**92.00**	47.80	63.25
H3/VDN DMD8	**78.15**	**77.68**	88.94	95.03	**98.07**	**92.00**	**54.21**	**68.07**
Synthetic: 5 s; $n = 25$ per configuration								
H3 50-step	**82.20**	**87.93**	93.34	**95.07**	98.11	**100.00**	61.28	65.08
LongVie2	80.24	87.02	**93.61**	94.82	97.37	**100.00**	57.48	60.10
LongVie AR2	79.61	86.30	92.46	94.86	96.97	**100.00**	55.69	61.42
H3/VDN DMD8	82.03	86.75	93.42	94.52	**98.24**	**100.00**	**61.73**	**65.90**
Synthetic: 14 s; $n = 25$ per configuration								
H3 50-step	80.59	82.85	91.61	94.52	97.69	**100.00**	59.21	67.88
LongVie2	78.47	81.55	91.26	94.81	97.29	**100.00**	56.23	59.69
LongVie AR2	78.78	82.85	91.30	**94.89**	96.98	**100.00**	55.78	61.47
H3/VDN DMD8	**81.38**	**83.28**	**92.32**	94.12	**98.05**	**100.00**	**62.72**	**67.94**

D Runtime, Resources, and Applications

D.1 Controlled DiT Timing

Primary timings cover synchronized model forward passes, including the H3/VDN backbone, Union branch, and FSDP/Ulysses communication. Each rank synchronizes local CUDA before and after each forward. We sum the maximum rank time at each of eight steps per window. Packing, Euler updates, history projection, depth preparation, VAE operations, validation, and output are excluded. Four fixed sources per canvas receive a full-window warmup and three hot repeats on isolated eight-H100 workers. The 32 total requests include warmups; reported hot means use 12 requests per canvas.

Table S16 DiT timing by window, means over 12 hot requests at each resolution. Full-continuation windows meet the 14-FPS throughput target at both resolutions. The last window delivers only 69 frames after overlap removal and padding trim. Step time is the mean of synchronized maximum-rank forward times.

Canvas	Window	New frames	DiT (s)	Step (s)	New FPS	RTF@24fps
960×544	Initial	345	21.144	2.643	16.346	1.471
960×544	Full continuation	306	21.175	2.647	14.472	1.661
960×544	Cropped tail	69	10.455	1.307	6.616	3.636
672×384	Initial	345	14.667	1.833	23.548	1.020
672×384	Full continuation	306	14.356	1.795	21.361	1.126
672×384	Cropped tail	69	8.571	1.071	8.072	2.981

Table S17 Whole-trajectory DiT throughput. Mean FPS averages each hot request's ratio. Pooled FPS divides all delivered frames by all elapsed DiT time; pooled steady FPS uses only full continuation windows.

Canvas	Total (s)	Mean FPS	RTF@24fps	Pooled FPS	Pooled steady FPS
960×544	52.774	13.665	1.759	13.643	14.451
672×384	37.594	19.183	1.253	19.152	21.315

The 345/306/69 new-frame counts sum to the 720-frame delivery in the 24-FPS evaluation stream. A complete continuation has 12.75 seconds of evaluation playback and a 1.59375-second average forward budget at eight evaluations. The 672×384 and 960×544 full-continuation paths reach 21.361 and 14.472 new FPS,

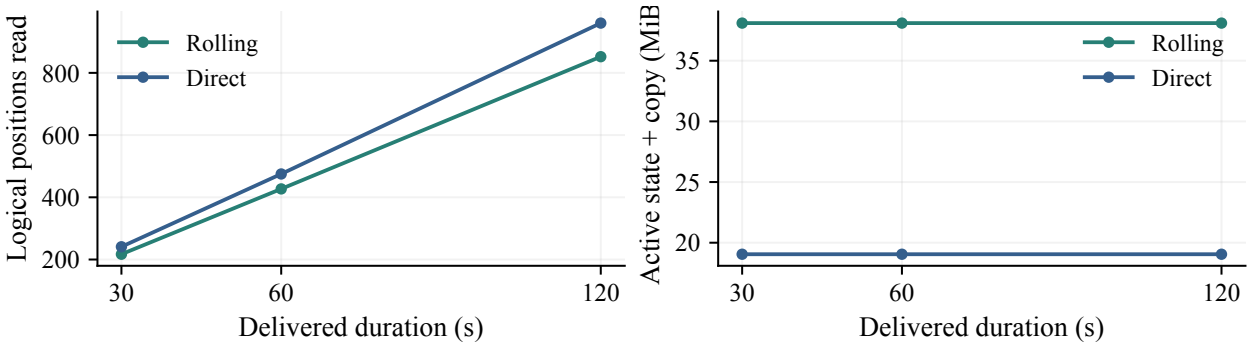


Figure S2 Logical depth reads and retained state plus consumer copy at 960 × 544. The resident global source is additional and grows with duration. Rolling reduces read positions but retains an extra owned state tensor.

respectively, meeting the 14-FPS throughput target at both resolutions. Timing runs reproduce the final latents of the corresponding requests.

D.2 Measured Depth-State and Model Resources

Resource probes use one warmup and one measured full first window per canvas, with eight actual forwards on each of eight ranks. Depth rows, carrier, and Union hints are measured from tensors. Local parameter storage and allocator peaks are measured separately. Incremental peaks include temporary activations and FSDP gather buffers; these components cannot be additively separated from this probe. The video VAE is not resident in the probe, so these measurements describe the measured model and conditioning state rather than a full-renderer peak.

Table S18 Actual first-window tensors and allocator increments. Hints are per rank; model storage and allocation increase take the maximum rank. Units distinguish MiB tensors from GiB model/allocator quantities.

Canvas	Depth (MiB)	Carrier (MiB)	Hints (MiB)	Model (GiB)	Increase (GiB)
960 × 544	19.050	39.276	347.710	9.025	7.034
672 × 384	9.413	19.407	177.393	9.025	5.090

The CPU benchmark uses actual FP32 depth, two PyTorch threads, one warmup, and 100 measurements per case, with the same validation/copy semantics for both readers. Four 30-second sources at each canvas and two longer geometries give ten cases. Table S19 averages per-source switch means and, separately, per-source p95 values; the latter is not a pooled p95 and differs from the DiT-window statistic.

Table S19 Resident CPU source switching, four 30-second sources per canvas, 100 repeats per source. State + copy excludes the global source.

Canvas	Reader	Mean switch (ms)	Mean source p95 (ms)	State + copy (MiB)
960 × 544	Rolling	11.836	18.088	38.101
960 × 544	Direct	7.377	11.509	19.050
672 × 384	Rolling	3.502	6.026	18.826
672 × 384	Direct	2.807	4.253	9.413

At 960 × 544, rolling requests 217/427/852 unique positions for 30/60/120-second deliveries; direct reads request 241/475/960. These counts measure logical source requests. Rolling concatenates retained overlap, while direct slicing accesses the resident source; Table S19 records the corresponding switch costs. Global depth, global noise, and committed outputs scale with duration and are accounted for separately. For 30 seconds at 960 × 544, global depth and noise occupy approximately 40.529 and 41.015 MiB, additional to active state and the carrier.

D.3 Long-Horizon Results

Table S20 Long-video aggregates. The 17 sixty-second sources use existing validation/test splits; the thirteen 120-second sources use an existing training split and are stress tests. Source sets differ.

Seconds	n	PSNR↑	SSIM↑	LPIPS↓	Rank↑	Edge↑	NMAE↓
60	17	14.4542	0.3198	0.4903	0.9114	0.6928	0.0999
120	13	14.4277	0.3358	0.5464	0.9353	0.6809	0.1771

The 60-second PSNR confidence interval is $[13.5573, 15.3446]$ dB; its LPIPS interval is $[0.43966, 0.54186]$. For 120 seconds these intervals are $[13.4260, 15.4057]$ dB and $[0.50169, 0.59344]$. Fixed-scale NMAE is 0.09993 ($[0.06854, 0.13587]$) and 0.17714 ($[0.05376, 0.38348]$), respectively.

Over the first to last ten-second block, 60-second PSNR changes from 18.257 to 13.136 dB, and fixed-scale NMAE from 0.0580 to 0.1428. The 120-second values change from 18.665 to 13.307 dB and from 0.0903 to 0.2248. Figure 5 characterizes these time-dependent measurements with source-group confidence bands while keeping geometric calibration fixed throughout each trajectory.

D.4 On-Demand Geometry, Appearance, and Control

Three 30-second, 672×384 procedural trajectories share one small PyBullet room. Warm/cool conditions use the same camera path and geometry with different reference materials; their depth is bitwise identical. The camera returns to its initial viewpoint at frame 480 (20 seconds). The control branch is enabled only when requesting the second window, with changes beginning at frame 357; the initial request has already read geometry through frame 356.

The cached cases run before the online cases; the first cached warm case may include cold compilation. Online execution requests new engine depth, encodes canonical blocks, and delivers RGB intervals $[0, 340)$, $[340, 646)$, and $[646, 720)$. Every newly acquired depth interval matches the cached oracle, and all three final video-latent outputs match bitwise. The recorded memory includes the retained global oracle, noise, and output latents.

Table S21 Appearance, native inverse-depth agreement, and revisit error for the three online conditions. Revisit columns compare frame 480 with frame zero using LPIPS; smaller is better. These are individual cases, not replicated population estimates.

Condition	LPIPS↓	Rank↑	Edge↑	NMAE↓	Revisit pred.	Revisit GT
Warm	0.4558	0.8915	0.5280	0.1170	0.4769	0.0056
Cool	0.2473	0.9115	0.5686	0.0986	0.2635	0.0064
Control	0.4561	0.8909	0.5253	0.1170	0.4668	0.0118

Warm/cool generated RGB differs by a mean 0.19322 on $[0, 1]$, versus 0.14456 for engine images; mean palette-change error is 0.03171. The control pair's initial-window latents are identical, while suffix latent MAE is 0.039076. The shared generator therefore responds to appearance and geometric-branch changes while preserving committed history. Table S21 reports absolute appearance, geometry, and revisit measurements for each case.

D.5 Window and Overlap Configurations

Four 30-second scenes—architecture, coast, gym, and office—compare overlaps 39/22/5 at 672×384. The corresponding windows are 345/328/311 frames with 12/7/2 history latents, all with stride 306. First-window content and commitment points change, unlike the fixed-window comparisons above.

The full-video ranking favors 39-frame overlap; the shared continuation $[345, 720)$ slightly favors shorter overlaps. A separate two-scene 60-second comparison also showed no consistent PSNR/SSIM advantage from history. These are configuration observations, not isolated estimates of the history constraint.

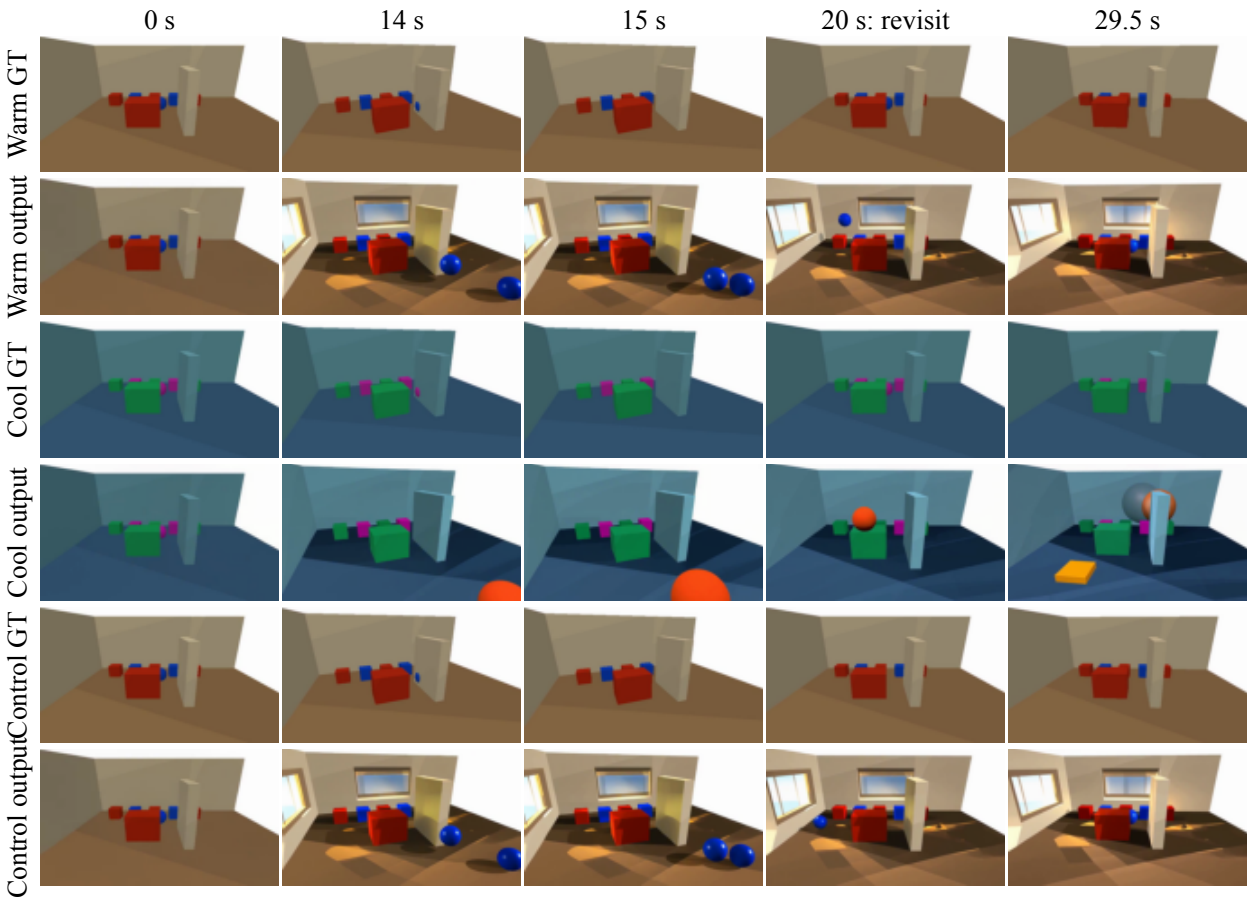


Figure S3 One shared renderer for appearance and geometric control. Engine RGB and generated frames for warm/cool references and a control branch, at fixed times. Warm/cool conditions share depth; control changes begin at frame 357 and the camera revisits at 20 seconds.

Table S22 Joint window-and-overlap configurations. PSNR averages per-frame dB; SSIM uses an 11×11 Gaussian window with $\sigma = 1.5$. These definitions differ from the primary evaluation.

Overlap	Window	History	Full PSNR	Full SSIM	Common-cont. PSNR
39	345	12	13.842738	0.341540	12.788645
22	328	7	13.683256	0.332978	12.861136
5	311	2	13.685815	0.336063	12.866656

D.6 Qualitative Selection and Paired Visualizations

Figure 4 uses the first selected ablation source from each category, independently of quality scores. The same four sources appear in the paired seam figures. Each row shows the same absolute timestamps across methods, and the reference is the actual conditioning image.

The figures pair source RGB with identical-time generated observations, complementing the source-fidelity, geometric, and temporal metrics. The simulation and synthetic rows display layout and appearance across both continuation boundaries. The online figure further varies reference appearance and future geometry within one shared model.

E AI Use and Reproducibility Statements

E.1 AI use statement

Generative AI assisted manuscript organization, drafting, editing, equation and pseudocode formatting, and preparation of tables and figures. Quantitative results and qualitative examples derive from completed model runs; method diagrams are illustrations. The authors remain responsible for the experimental results and final manuscript.

E.2 Reproducibility statement

Equations 4–11 and Algorithm 1 specify inference. Appendices A–C provide state proofs, model components, VAE phase, depth conventions, valid masks, source-selection rules, paired controls, and bootstrap statistics. Appendix D specifies the hardware, warmup and repetition settings, and measurement scopes for DiT and resource measurements; the online subsection reports source-interface behavior and control equivalence. The supplementary tables report aggregate results and uncertainty under these protocols.